Introduction

The Problem

Educational attainment varies considerably between secondary schools in England. Understanding how school and pupil characteristics relate to these differences is an important problem, as it may help identify patterns associated with stronger or weaker academic performance. In this project, we set out to predict the average Attainment 8 score of English secondary schools using information about the schools and their pupils. Attainment 8 measures pupils' performance across eight qualifications, making it a useful indicator of academic achievement at GCSE level.

Our dataset contains a range of potential predictors, including school size, admission type, pupil demographics, special educational needs and the proportion of disadvantaged pupils. We investigated whether these characteristics can be used to predict a school's average Attainment 8 score accurately.

Within our group we studied and compared three methods:

- Linear regression
- Random Forest Regression
- Gradient Boosting Regression

We compared predictive accuracy of these models using metrics such as RMSE, MAE and R^2. We wanted to determine which approach produces the most accurate predictions, while considering model complexity, overfitting and interpretability.

Obtaining the Data

We collected our data from the Department for Education's datasets on GOV.UK, covering the academic years 2023/24, 2024/25 and 2025/26. The hyperlink for the data:

https://explore-education-statistics.service.gov.uk/find-statistics/key-stage-4-performance/2025-26#content

Since the information was spread across several CSV files, we used AI to help clean and combine the datasets. This involved matching schools across different files and years, selecting relevant variables and dealing with missing values. The resulting dataset, school_level_master_age_split.csv, contained 17,264 observations and 78 variables.

Before splitting the data, we removed observations with missing values in the variables selected for modelling, leaving us with 9,842 observations. We then used an 80/20 training and test split, giving us

7,873 training observations and 1,969 test observations. The same split was used across all three models to allow for a fair comparison.

There are some limitations to this approach. Removing observations with missing values could make the sample less representative of all schools. Also, the same school can appear in both the training and test sets for different academic years. This means the models may have already encountered information about a school, potentially making the test results more optimistic than they would be for completely unseen schools.

The datasets can be found under Report → 1_Introduction → 1_Data_Google_Drive, with the splitting code in 1_Splitting_Data.ipynb. We also included the saved training and test datasets so that our results can be reproduced.

Understanding the Data

The Target Variable

Our target variable for the data will be 'attainment8_average', this is a measure of the average student's GCSE performance at each school in the data set. Attainment 8 is calculated as the sum of their best 8 grades across three main elements:

- English and Maths (double-weighted)
    - Maths is always worth double
    - English is worth double if both English Language and English Literature are taken; if so then the higher grade of the two is worth double, and then the other grade can be used in the open slot.
- EBacc Subjects (three slots)
  Takes the top three grades from the following subjects
    - Separate Sciences (Physics, Chemistry, Biology)
    - Combined Science (can take up to two slots with the average grade)
    - Computer Science
    - History
    - Geography
    - Languages
- Open Subjects
  Takes the top three grades from any other GCSE subjects which weren't used as one of the English and Maths or EBacc slots. Can also include certain non-GCSE Level 2 qualifications.

The attainment 8 score is therefore has a minimum of 0 and a maximum of 90 (due to maths and english counting for double)

The Explanatory Variables

The remaining columns in the data are the exploratory variables we will use in our models. These are the following:

- School_urn - Unique Reference Number (URN) for each school.
- Time_period - Academic year the data relates to (e.g. 202526 means 2025/26).
- School_name - Name of the school.
- School_laestab - Local Authority and Establishment number identifying each school.
- La_name - Name of the Local Authority in which the school is located.
- New_la_code - Government code which identifies the Local Authority.
- Establishment_type_group - Type of school (e.g. academy or independent school).
- Pcon_code - Parliamentary constituency code for the school's location.
- Pcon_name - Name of the parliamentary constituency the school is in.
- School_admission_type - Type of admissions policy (e.g. selective or non-selective).
- School_religious_character_detailed - Religious affiliation of the school (where applicable).
- Sex_policy - Whether the school is boys-only, girls-only or mixed.
- Age_range - Age range of pupils attending the school.
- Min_age - Minimum age of pupils at the school.
- Max_age - Maximum age of pupils at the school.
- Allks_pupil_count - Total number of pupils across all Key Stages.
- Allks_boys_count - Total number of boys across all Key Stages.
- Allks_girls_count - Total number of girls across all Key Stages.
- Endks4_pupil_count - Number of pupils at the end of Key Stage 4.
- Sen_pupil_count - Number of pupils identified as having special educational needs (SEN).
- Sen_pupil_percent - Percentage of pupils identified as having SEN.
- Sen_with_ehcp_pupil_count - Number of pupils with SEN who have an Education, Health and Care Plan (EHCP).
- Sen_with_ehcp_pupil_percent - Percentage of pupils with an EHCP.
- Sen_no_ehcp_pupil_count - Number of pupils with SEN but without an EHCP.
- Sen_no_ehcp_pupil_percent - Percentage of pupils with SEN but without an EHCP.
- *Attainment8eng_average - Average Attainment 8 points from English.*
- *Attainment8mat_average - Average Attainment 8 points from Mathematics.*
- *Attainment8ebacc_average - Average Attainment 8 points from EBacc subjects.*
- *Attainment8open_average - Average Attainment 8 points from the remaining open subject slots.*
- *Pupil_count - Number of pupils included in the GCSE performance measures.*
- *Engmath_94_percent - Percentage of pupils achieving grade 4 or above in both English and Mathematics.*
- *Engmath_95_percent - Percentage of pupils achieving grade 5 or above in both English and Mathematics.*
- *Engmath_97_percent - Percentage of pupils achieving grade 7 or above in both English and Mathematics.*
- *Engmath_94_total - Number of pupils achieving grade 4 or above in both subjects.*
- *Engmath_95_total - Number of pupils achieving grade 5 or above in both subjects.*
- *Engmath_97_total - Number of pupils achieving grade 7 or above in both subjects.*

- *Gcse_91_percent - Percentage of pupils achieving grade 9–1 in at least one GCSE.*
- Gcse_entries_average - Average number of GCSE entries per pupil.
- Qual_entries_average - Average number of qualifications entered per pupil, including relevant non-GCSE qualifications.
- Sci_triple_entering_percent - Percentage of pupils entered for GCSE Triple Science.
- Lan_multiple_entering_percent - Percentage of pupils entered for multiple language qualifications.

- We then have that for each of the following prefixes ks4_boys_ , ks4_girls_, ks4_disadvantaged_ , ks4_not_known_disadvantaged_, ks4_first_language_other_than_english_ and ks4_non_mobile_ there are the following explanatory variables that start with each of the prefixes:
    - Pupil_count
    - Pupil_percent
    - *Attainment8_average*
    - *Engmath_94_percent*
    - *Engmath_95_percent*
    - *Progress8_average*

  The prefixes are defined as:

    - Ks4_boys_ - For boys in KS4.
    - Ks4_girls_ - For girls in KS4.
    - Ks4_disadvantaged_ - For pupils in KS4 classified as disadvantaged. A student is considered disadvantaged if:
        - They have been eligible for free school meals at any point in the last 6 years.
        - They have at some point been in the care of a local authority.
    - Ks4_not_known_disadvantaged_ - For pupils in KS4 whose disadvantages status is not known.
    - Ks4_first_language_other_than_english_ - For KS4 students whose first language is not english.
    - Ks4_non_mobile_ - For KS4 students who haven't moved school during KS4.

The columns written in italic are columns that include information about previous academic performance and therefore will be left out of the data in most of our actual analysis.